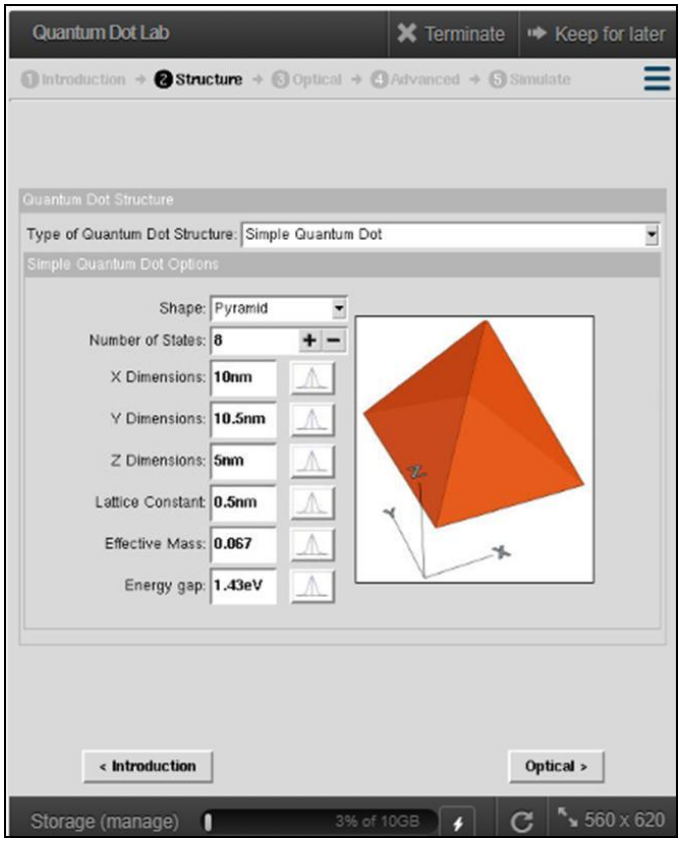
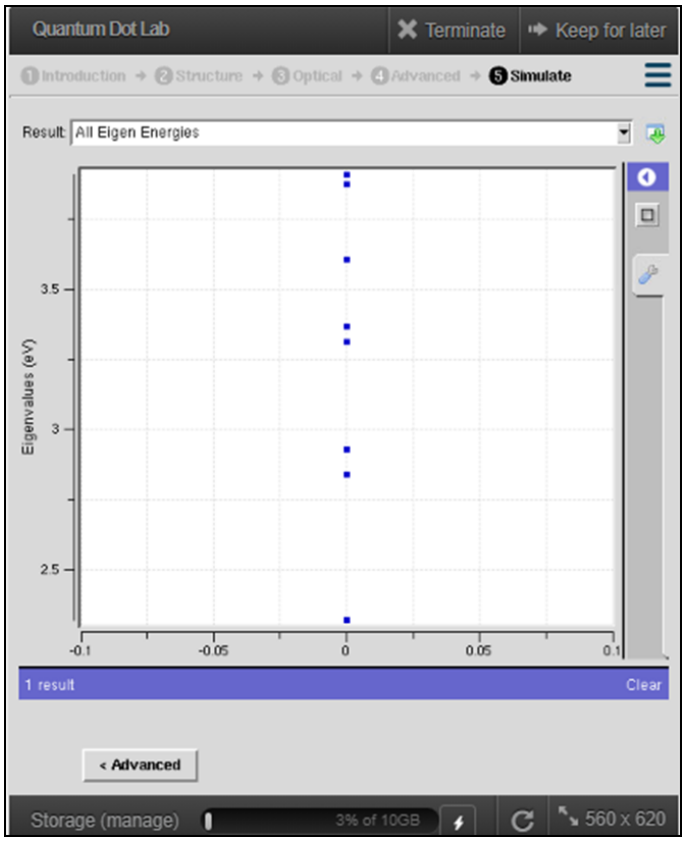
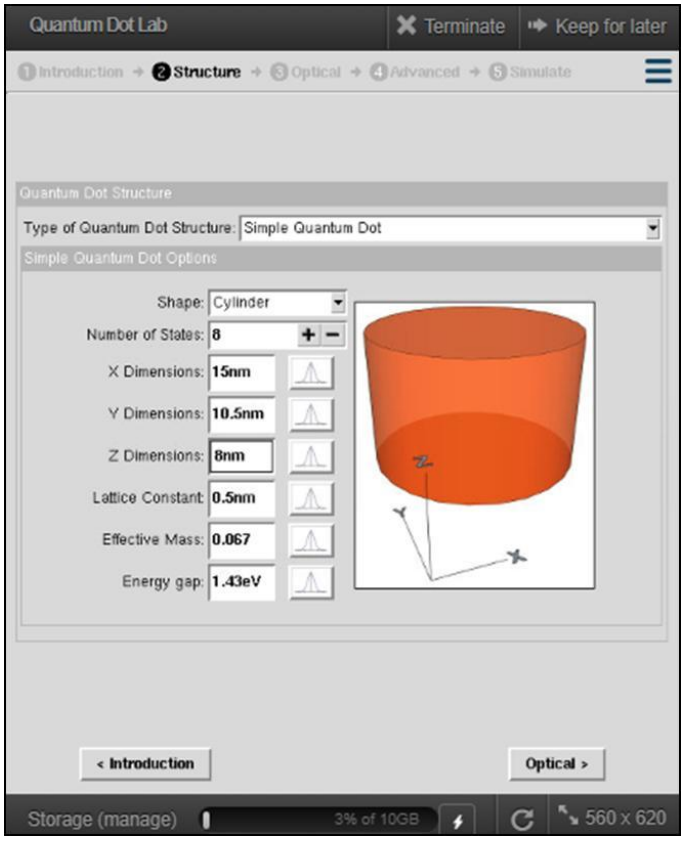
ECA5701-NANOELECTRONICS

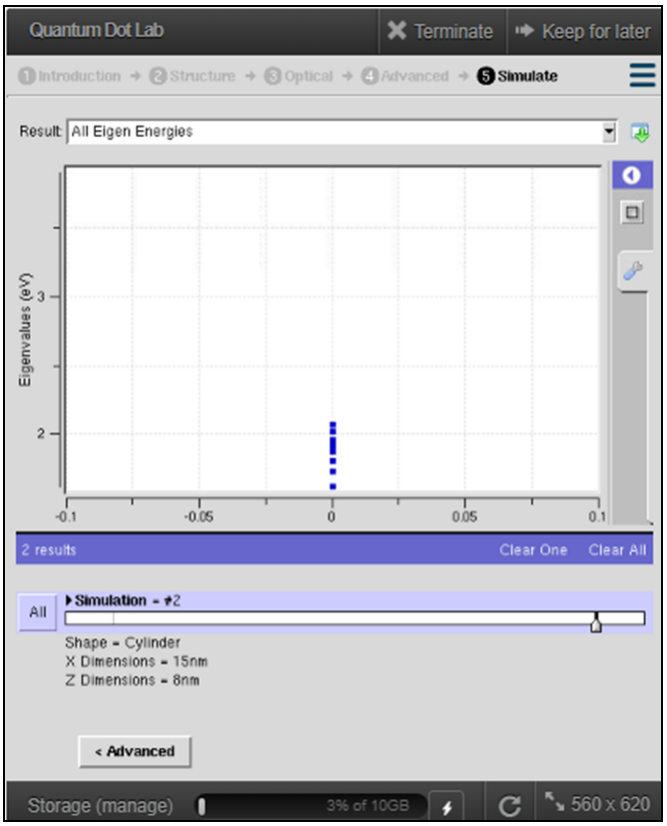
JAYANTH R
192512342

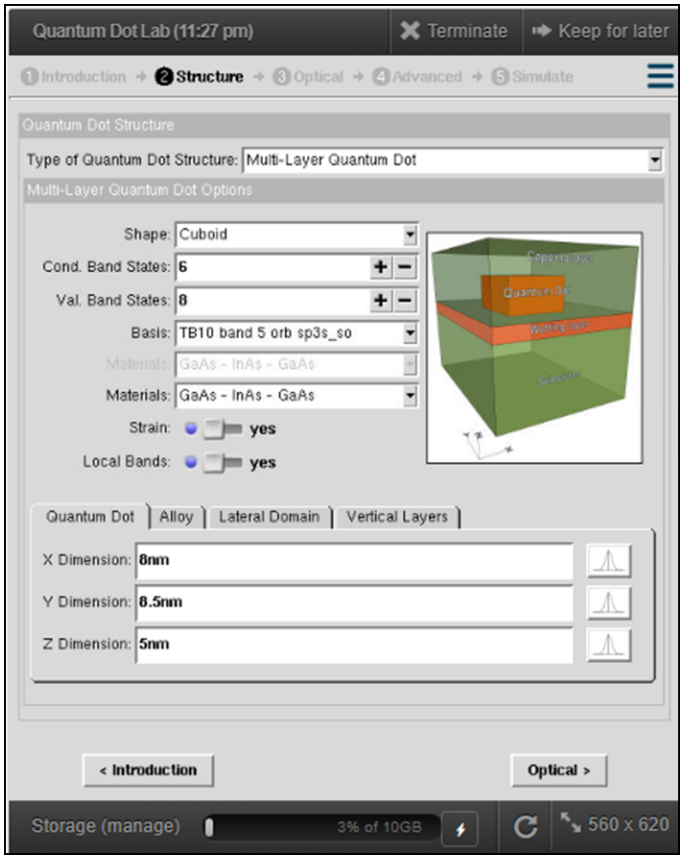
Exp. 3 – Energy States of Quantum Dot using NanoHub

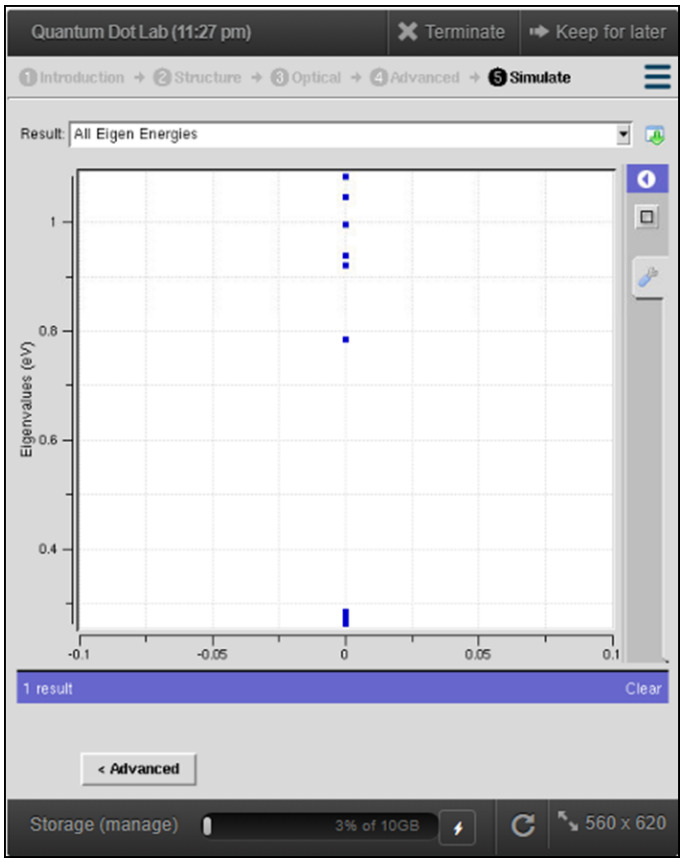





Quantum Dot Lab
Terminate
Keep for later
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Simple Quantum Dot
Simple Quantum Dot Options
Shape: Cylinder
Number of States: 8
X Dimensions: 15nm
Y Dimensions: 10.5nm
Z Dimensions: 8nm
Lattice Constant: 0.5nm
Effective Mass: 0.067
Energy gap: 1.43eV
< Introduction
Optical >
Storage (manage)
3% of 10GB
560 x 620


Quantum Dot Lab
Terminate
Keep for later
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
2 results
Clear One
Clear All
All
Simulation - ★2
Shape = Cylinder
X Dimensions = 15nm
Z Dimensions = 8nm
< Advanced
Storage (manage)
3% of 10GB
560 x 620


Quantum Dot Lab (11:27 pm)
Terminate
Keep for later
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Multi-Layer Quantum Dot
Multi-Layer Quantum Dot Options
Shape: Cuboid
Cond. Band States: 6
Val. Band States: 8
Basis: TB10 band 5 orb sp3s_so
Materials: GaAs - InAs - GaAs
Strain: yes
Local Bands: yes
Quantum Dot
Alloy
Lateral Domain
Vertical Layers
X Dimension: 8nm
Y Dimension: 8.5nm
Z Dimension: 5nm
< Introduction
Optical >
Storage (manage)
3% of 10GB
560 x 620


Quantum Dot Lab (11:27 pm)
Terminate
Keep for later
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
1 result
Clear
< Advanced
Storage (manage)
3% of 10GB
560 x 620